RUDRA PRASAD RATH

CYBERSECURITY & AI SECURITY ENGINEER

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PROFESSIONAL SUMMARY

Cybersecurity & AI Security Engineer with practical experience in defensive software architecture, automated intrusion detection, and adversarial vulnerability assessment. Experienced in engineering custom Intrusion Detection and Prevention Systems (IDPS) in Python/Flask, conducting web and network penetration testing using Kali Linux, Burp Suite, and Wireshark, and researching LLM threat mitigation (OWASP Top 10 for LLMs, prompt injection defense). Adept at bridging AI/ML foundations with cybersecurity engineering to automate threat mitigation and enforce security-by-design principles across web, API, and intelligent systems.

CORE TECHNICAL SKILLS

AI & GenAI Security: Adversarial Machine Learning, OWASP Top 10 for LLMs, Prompt Injection & Jailbreak Defense, MITRE ATLAS Threat Modeling, NIST AI RMF, RAG Security Guardrails, Model Input Validation.

Cybersecurity & Pen-Testing: Intrusion Detection & Prevention (IDS/IPS), Web App Security (OWASP Top 10: SQLi, XSS, CSRF), Network Packet Analysis (Wireshark PCAP), Port Scanning & Reconnaissance (Nmap), Metasploit, Burp Suite.

Programming & Automation: Python (Advanced Security Tooling, Regex Engines), Java, C, SQL, Bash Scripting, REST API Hardening, JSON Data Normalization, Git/GitHub.

Libraries, Databases & OS: Flask, NumPy, Pandas, Scikit-Learn [NEEDS INPUT], SQLite, MySQL, Linux (Debian/Ubuntu Hardening).

Frameworks & Standards: MITRE ATT&CK, MITRE ATLAS, NIST Cybersecurity Framework (CSF), Defense-in-Depth, Zero Trust Principles.

PROFESSIONAL EXPERIENCE

Cybersecurity Intern | UNLOX (Remote) — Apr 2026 – May 2026

- **Engineered automated vulnerability scanning workflows** across Linux and TCP/IP networks using Nmap and Wireshark, identifying open attack vectors and misconfigurations.
- **Conducted systematic web application security testing** with Burp Suite and Kali Linux, analyzing OWASP Top 10 vectors including input sanitization defects and auth bypasses.
- **Formulated comprehensive technical remediation reports** mapping exploit pathways to CVSS-aligned risk severity ratings, delivering actionable hardening steps.
- **Simulated adversarial attack exercises** in controlled laboratory environments to validate the resilience of defensive firewall controls and intrusion detection rules.
- Identified and documented **[NEEDS INPUT: e.g., 20+]** security vulnerabilities, accelerating remediation turnaround time for staging deployments.

SELECTED AI & CYBERSECURITY PROJECTS

Web-Based Intrusion Detection & Prevention System (IDPS) — Python, Flask, SQLite, Regex | 2026

- **Architected an automated web defense engine** that intercepts, logs, and mitigates SQL injection, Cross-Site Scripting (XSS), and brute-force authentication attacks in real time.
- **Implemented stateful IP-based rate limiting** (60 req/min thresholds) and automated temporary IP null-routing, dropping malicious traffic before backend query execution.
- **Built an administrative telemetry dashboard** streaming real-time security events, sanitizing user inputs, and maintaining persistent audit logs for SOC monitoring.

OSINT Telephony Intelligence & Geolocation Engine — Python, REST APIs, JSON, Geocoding | 2026

- **Engineered a 10-module Python threat intelligence tool** designed to automate Open-Source Intelligence (OSINT) reconnaissance on global telephony indicators.
- **Integrated third-party geocoding and telecom routing REST APIs** to parse line validation, carrier identity, timezone metadata, and approximate coordinates.
- **Developed an automated visual rendering pipeline** translating normalized JSON telemetry into interactive Leaflet HTML maps for rapid SOC analyst triage.

Adversary Emulation & Penetration Testing Labs — Kali Linux, Metasploit, Nmap, Wireshark | Ongoing

- **Executed structured offensive security simulations** on Hack The Box lab environments targeting vulnerable Linux/Windows machines.
- **Conducted end-to-end exploit lifecycles**: service enumeration via Nmap, exploit delivery with Metasploit, binary privilege escalation, and lateral movement analysis.

EDUCATION

B.Tech in Computer Science & Engineering (AI/ML) — 2024 – 2028

NIST University, Berhampur, Odisha • CGPA: 7.3 / 10.0

Relevant Coursework: Machine Learning, Artificial Intelligence, Cybersecurity, Data Structures & Algorithms, Network Security, Database Management Systems.

CERTIFICATIONS & CREDENTIALS

Introduction to Cybersecurity — Cisco Networking Academy (Jun 2026) | Threat Analysis, Network Defense, Cyber Administration.

Hands-on Cybersecurity Labs — Hack The Box (2026) | Network Enumeration, Metasploit Exploitation, Packet Analysis.

LANGUAGES

English (Professional Working Proficiency), Hindi (Fluent), Odia (Native).